

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Contents

1	Monday 6 July 2026	2
1.1	Medium Differential	2
1.2	Hard Differential	2
2	Tuesday 7 July 2026	4
2.1	Medium Differential	4
2.2	Hard Differential	4
3	Wednesday 8 July 2026	6
3.1	Medium Differential	6
3.2	Hard Differential	6
4	Thursday 9 July 2026	8
4.1	Medium Differential	8
4.2	Hard Differential	8
5	Friday 10 July 2026	10
5.1	Medium Differential	10
5.2	Hard Differential	10

Monday 6 July 2026

1.1 Medium Differential

Simple Harmonic Motion

A simple pendulum on Planet Waifu consists of a particle of mass m attached to a light, inextensible string of length $l = 4$ m. The local acceleration due to gravity is $g = 100 \text{ m/s}^2$. The pendulum is released from its equilibrium position at $t = 0$ with an initial linear velocity of $v(0) = 10 \text{ m/s}$. Assuming the system allows for pure Simple Harmonic Motion find the linear displacement s along its arc after exactly 1 second.

Solution: For pure Simple Harmonic Motion, the differential equation governing linear displacement s is:

$$\frac{d^2s}{dt^2} + \omega^2 s = 0$$

where $\omega = \sqrt{\frac{g}{l}}$. Given $g = 100$ and $l = 4$:

$$\omega = \sqrt{\frac{100}{4}} = \sqrt{25} = 5 \text{ rad/s}$$

The general form of displacement is $s(t) = A \cos(5t) + B \sin(5t)$. Because it is released from its equilibrium position at $t = 0$, $s(0) = 0$:

$$s(0) = A(1) + B(0) = 0 \implies A = 0$$

Thus, $s(t) = B \sin(5t)$. Taking the derivative to find the velocity:

$$v(t) = s'(t) = 5B \cos(5t)$$

Apply the initial velocity condition $v(0) = 10$:

$$v(0) = 5B \cos(0) = 10 \implies 5B = 10 \implies B = 2$$

The specific equation for linear displacement is:

$$s(t) = 2 \sin(5t)$$

Evaluating at exactly 1 second ($t = 1$):

$$s(1) = 2 \sin(5)$$

1.2 Hard Differential

System of Differential Equations

Solve the system of differential equations:

$$\begin{cases} \frac{dx}{dt} + 2xy + 2x = -\frac{dy}{dt} - 2y \\ x \frac{dy}{dt} + y \frac{dx}{dt} = \frac{x+y}{2} \end{cases}$$

Knowing that $x(0) = 0$ and $y(0) = 1$, find the value of $x(2)y(2) + x(1) + y(1)$.

Solution: Rearrange the two equations by making sensible group substitutions. 1. The first equation can be rearranged to group derivatives and terms linearly:

$$(x' + y') + 2(x + y) = -2xy$$

2. The second equation represents the exact derivative of a product:

$$(xy)' = \frac{x + y}{2}$$

Let $u = x + y$ and $v = xy$. The system radically simplifies to:

$$u' + 2u = -2v$$

$$v' = \frac{u}{2} \implies u = 2v'$$

Substitute $u = 2v'$ into the first equation:

$$(2v')' + 2(2v') = -2v \implies 2v'' + 4v' + 2v = 0 \implies v'' + 2v' + v = 0$$

The characteristic equation is $(r + 1)^2 = 0$, yielding a repeated root $r = -1$.

$$v(t) = (C_1 + C_2 t)e^{-t}$$

Apply initial conditions $x(0) = 0$ and $y(0) = 1$: $u(0) = 0 + 1 = 1$ $v(0) = (0)(1) = 0$ Solving for constants: $v(0) = C_1 = 0 \implies v(t) = C_2 t e^{-t}$ $u(t) = 2v'(t) = 2C_2(1 - t)e^{-t}$ Using $u(0) = 1$, we get $2C_2(1) = 1 \implies C_2 = 1/2$. The solutions in terms of our dummy variables are:

$$v(t) = \frac{1}{2} t e^{-t} \quad \text{and} \quad u(t) = (1 - t)e^{-t}$$

We need to find the specific value of $x(2)y(2) + x(1) + y(1)$, which translates to $v(2) + u(1)$:

$$v(2) = \frac{1}{2}(2)e^{-2} = e^{-2}$$

$$u(1) = (1 - 1)e^{-1} = 0$$

$$v(2) + u(1) = e^{-2} + 0 = e^{-2}$$

Tuesday 7 July 2026

2.1 Medium Differential

Non-linear ODE (Bernoulli Type)

A function y satisfies:

$$\frac{xy'}{y} + xe^x + 1 = x^5 e^x y^4$$

Given that $y(1) = 1$, find $y(2)$.

Solution: First, rearrange the equation into a standard format by dividing by x and multiplying by y :

$$y' + \left(\frac{1}{x} + e^x\right)y = x^4 e^x y^5$$

This is a Bernoulli equation with $n = 5$. Let $v = y^{1-5} = y^{-4}$. The derivative is $v' = -4y^{-5}y'$. Multiply the rearranged equation by $-4y^{-5}$:

$$-4y^{-5}y' - 4\left(\frac{1}{x} + e^x\right)y^{-4} = -4x^4 e^x \implies v' - 4\left(\frac{1}{x} + e^x\right)v = -4x^4 e^x$$

This forms a first-order linear ODE for v . The integrating factor $I(x)$ is:

$$I(x) = \exp\left(-4 \int \left(\frac{1}{x} + e^x\right) dx\right) = \exp(-4 \ln x - 4e^x) = x^{-4} e^{-4e^x}$$

Multiplying both sides by $I(x)$ results in:

$$\left(vx^{-4}e^{-4e^x}\right)' = -4e^x e^{-4e^x}$$

Integrate both sides with respect to x :

$$vx^{-4}e^{-4e^x} = \int -4e^x e^{-4e^x} dx = e^{-4e^x} + C$$

Multiplying through by $x^4 e^{4e^x}$ provides $v(x)$:

$$v(x) = x^4 + Cx^4 e^{4e^x}$$

Substituting $v = y^{-4}$:

$$y^{-4} = x^4(1 + Ce^{4e^x})$$

Apply the initial condition $y(1) = 1$:

$$1^{-4} = 1^4(1 + Ce^{4e}) \implies 1 = 1 + Ce^{4e} \implies C = 0$$

Our final specific function is $y^{-4} = x^4 \implies y(x) = \frac{1}{x}$ (choosing the positive root since $y(1) > 0$). Evaluate at $x = 2$:

$$y(2) = \frac{1}{2}$$

2.2 Hard Differential

PDE and Laplace Transform

A homogeneous nano-rod of length $L = 6$ m is fixed at one end ($x = 0$) and rotates in a horizontal plane around a vertical axis. The temperature evolution $\theta(x, t)$ is modeled by the PDE:

$$\frac{\partial^2 \theta}{\partial x^2} + \beta^2 x = \frac{1}{a^2} \frac{\partial \theta}{\partial t}$$

Subject to the initial and boundary conditions: $\theta(x, 0) = 0$, $\theta(0, t) = 0$, $\frac{\partial \theta}{\partial x}(L, t) = 0$
Using the Laplace transform method, determine the steady-state distribution $\theta(x, \infty)$ and evaluate $\theta(3, \infty)$ for $\beta = 2$.

Solution: Taking the Laplace Transform in t with zero initial conditions yields the ODE in space:

$$\frac{d^2 \Theta(x, s)}{dx^2} - \frac{s}{a^2} \Theta(x, s) = -\frac{\beta^2 x}{s}$$

The general solution to this non-homogeneous ODE is:

$$\Theta(x, s) = A(s) \cosh\left(\frac{\sqrt{s}}{a} x\right) + B(s) \sinh\left(\frac{\sqrt{s}}{a} x\right) + \frac{a^2 \beta^2 x}{s^2}$$

Applying $\Theta(0, s) = 0$ gives $A(s) = 0$. Applying $\frac{\partial \Theta}{\partial x}(L, s) = 0$ gives:

$$B(s) \frac{\sqrt{s}}{a} \cosh\left(\frac{\sqrt{s} L}{a}\right) + \frac{a^2 \beta^2}{s^2} = 0 \implies B(s) = -\frac{a^3 \beta^2}{s^{5/2} \cosh(\sqrt{s} L / a)}$$

We invoke the Final Value Theorem to find the steady-state profile: $\theta(x, \infty) = \lim_{s \rightarrow 0} s \Theta(x, s)$.

$$s \Theta(x, s) = -\frac{a^3 \beta^2}{s^{3/2} \cosh(\sqrt{s} L / a)} \sinh\left(\frac{\sqrt{s}}{a} x\right) + \frac{a^2 \beta^2 x}{s}$$

Taking the limit $s \rightarrow 0$ using small argument approximations $\sinh(z) \approx z + z^3/6$ and $1/\cosh(z) \approx 1 - z^2/2$:

$$\lim_{s \rightarrow 0} s \Theta(x, s) = -\frac{a^3 \beta^2}{s^{3/2}} \left(1 - \frac{s L^2}{2 a^2}\right) \left(\frac{\sqrt{s} x}{a} + \frac{s^{3/2} x^3}{6 a^3}\right) + \frac{a^2 \beta^2 x}{s}$$

Expanding the terms, the $1/s$ singularities perfectly cancel out, leaving:

$$\theta(x, \infty) = \frac{\beta^2 L^2}{2} x - \frac{\beta^2}{6} x^3$$

Evaluating $\theta(3, \infty)$ for $\beta = 2$ and $L = 6$:

$$\theta(3, \infty) = \frac{4(36)}{2}(3) - \frac{4}{6}(27) = 216 - 18 = 198$$

Wednesday 8 July 2026

3.1 Medium Differential

Signal Speed Differential Equation

A neural signal travelling through an artificial brainstem has speed $v(t)$ (in m/s) after t milliseconds. The speed satisfies the differential equation:

$$v'' - 4v' + 4v = te^{2t}$$

Given the initial conditions $v(0) = 3$ and $v'(0) = 10$, find the exact speed of the signal at $t = \ln(2)$.

Solution: We use the exponential shift method. Let $v(t) = u(t)e^{2t}$. We find the derivatives: $v' = (u' + 2u)e^{2t}$ $v'' = (u'' + 4u' + 4u)e^{2t}$ Substitute these into the left-hand side of the ODE:

$$(u'' + 4u' + 4u - 4(u' + 2u) + 4u)e^{2t} = u''e^{2t}$$

Equating to the right-hand side, $u''e^{2t} = te^{2t} \implies u'' = t$. Integrating twice:

$$u'(t) = \frac{t^2}{2} + C_1$$

$$u(t) = \frac{t^3}{6} + C_1t + C_2$$

Thus, the general solution for speed is:

$$v(t) = \left(\frac{t^3}{6} + C_1t + C_2 \right) e^{2t}$$

Using initial condition $v(0) = 3$:

$$v(0) = C_2 = 3$$

Taking the first derivative of $v(t)$:

$$v'(t) = \left(\frac{t^2}{2} + C_1 \right) e^{2t} + 2 \left(\frac{t^3}{6} + C_1t + 3 \right) e^{2t}$$

Using initial condition $v'(0) = 10$:

$$v'(0) = C_1 + 2(3) = 10 \implies C_1 = 4$$

The specific solution is:

$$v(t) = \left(\frac{t^3}{6} + 4t + 3 \right) e^{2t}$$

Evaluate at $t = \ln(2)$. Note that $e^{2\ln(2)} = e^{\ln(4)} = 4$.

$$v(\ln 2) = \left(\frac{(\ln 2)^3}{6} + 4\ln 2 + 3 \right) \times 4 = \frac{2}{3}(\ln 2)^3 + 16\ln 2 + 12$$

3.2 Hard Differential

Second-Order ODE via Variation of Parameters

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = \frac{e^x}{x^2 + 1}$$

Given $y(0) = 0, y'(0) = 1$, find $y(1)$.

Solution: First, solve the homogeneous equation $y'' - 2y' + y = 0$. The characteristic equation is $r^2 - 2r + 1 = 0 \implies (r - 1)^2 = 0$. The complementary solution is $y_c(x) = C_1e^x + C_2xe^x$, giving fundamental solutions $y_1 = e^x$ and $y_2 = xe^x$. The Wronskian is:

$$W(y_1, y_2) = e^x(e^x + xe^x) - (xe^x)(e^x) = e^{2x}$$

We apply Variation of Parameters to find a particular solution $y_p = u_1y_1 + u_2y_2$:

$$u_1' = \frac{-y_2f(x)}{W} = \frac{-xe^x \cdot \frac{e^x}{x^2+1}}{e^{2x}} = -\frac{x}{x^2+1} \implies u_1 = -\frac{1}{2}\ln(x^2+1)$$

$$u_2' = \frac{y_1f(x)}{W} = \frac{e^x \cdot \frac{e^x}{x^2+1}}{e^{2x}} = \frac{1}{x^2+1} \implies u_2 = \arctan(x)$$

The general solution is:

$$y(x) = C_1e^x + C_2xe^x - \frac{1}{2}e^x \ln(x^2+1) + xe^x \arctan(x)$$

Now apply the initial conditions. For $y(0) = 0$:

$$y(0) = C_1 - 0 + 0 = 0 \implies C_1 = 0$$

Differentiating $y(x)$:

$$y'(x) = C_2(e^x + xe^x) - \frac{1}{2}e^x \ln(x^2+1) - \frac{xe^x}{x^2+1} + (e^x + xe^x) \arctan(x) + \frac{xe^x}{x^2+1}$$

For $y'(0) = 1$:

$$y'(0) = C_2(1) - 0 - 0 + 0 + 0 = 1 \implies C_2 = 1$$

The specific solution is:

$$y(x) = xe^x - \frac{1}{2}e^x \ln(x^2+1) + xe^x \arctan(x)$$

Evaluate at $x = 1$:

$$y(1) = e - \frac{1}{2}e \ln(2) + e \arctan(1) = e \left(1 - \frac{\ln(2)}{2} + \frac{\pi}{4} \right)$$

Thursday 9 July 2026

4.1 Medium Differential

Logistic Population Growth

At any time t (in years), the rate of growth of the population N of deer in a state park is proportional to the product of N and $L - N$ where $L = 500$ is the maximum number of deer the park can maintain. In the beginning, the park has 100 deer, and after 4 years, the number of deer has increased to 200. After how many years (to 2dp) will the deer population be 350?

Solution: This is governed by the logistic differential equation $\frac{dN}{dt} = kN(L - N)$. The standard solution is:

$$N(t) = \frac{L}{1 + Ce^{-Lkt}}$$

Here $L = 500$. Let $r = 500k$, making the equation $N(t) = \frac{500}{1 + Ce^{-rt}}$. Apply the initial condition $N(0) = 100$:

$$100 = \frac{500}{1 + C} \implies 1 + C = 5 \implies C = 4$$

So, $N(t) = \frac{500}{1 + 4e^{-rt}}$. Apply the condition $N(4) = 200$:

$$200 = \frac{500}{1 + 4e^{-4r}} \implies 1 + 4e^{-4r} = 2.5 \implies 4e^{-4r} = 1.5 \implies e^{-4r} = \frac{3}{8}$$

We want to find t when $N(t) = 350$:

$$350 = \frac{500}{1 + 4e^{-rt}} \implies 1 + 4e^{-rt} = \frac{500}{350} = \frac{10}{7}$$

$$4e^{-rt} = \frac{3}{7} \implies e^{-rt} = \frac{3}{28}$$

Since $e^{-4r} = 3/8$, we have $e^{-r} = (3/8)^{1/4}$. Substituting this into our expression for t :

$$\left(\frac{3}{8}\right)^{\frac{t}{4}} = \frac{3}{28}$$

Take the natural logarithm of both sides:

$$\frac{t}{4} \ln\left(\frac{3}{8}\right) = \ln\left(\frac{3}{28}\right) \implies t = 4 \frac{\ln(3/28)}{\ln(3/8)} = 4 \frac{\ln(28/3)}{\ln(8/3)}$$

Numerically calculating this value:

$$t \approx 4 \times \frac{2.23359}{0.98083} \approx 4 \times 2.2772 = 9.1088$$

Rounding to two decimal places, the population reaches 350 after 9.11 years.

4.2 Hard Differential

Trigonometric Differential Equation

Assume the following relationship for $x > 0$:

$$\sin\left(\frac{dy}{dx}\right) = \sin(y)$$

Given the boundary conditions $y(0) = 8\pi$ and $y(\infty) = 5\pi$, determine the exact value of $y(1)$.

Solution: The equation $\sin(y') = \sin(y)$ implies that the arguments are related by either $y' = y + 2k\pi$ or $y' = (2k + 1)\pi - y$, where $k \in \mathbb{Z}$. Let's analyze the first case: $y' - y = 2k\pi$. The solution is of the form $y(x) = Ae^x - 2k\pi$. Because $x \rightarrow \infty$, $e^x \rightarrow \infty$, so to satisfy the bounded condition $y(\infty) = 5\pi$, we must have $A = 0$, leading to a constant solution $y = -2k\pi$. But $y(0) = 8\pi \neq 5\pi$, so this case is impossible.

We move to the second case: $y' = (2k + 1)\pi - y$. Let $C = (2k + 1)\pi$.

$$y' + y = C$$

This is a linear ODE with integrating factor e^x . The solution is:

$$y(x) = Ae^{-x} + C$$

Using the boundary condition at infinity:

$$y(\infty) = A(0) + C = 5\pi \implies C = 5\pi$$

Since $C = (2k + 1)\pi$, this is perfectly valid for $k = 2$. Now, apply the initial boundary condition at $x = 0$:

$$y(0) = A + 5\pi = 8\pi \implies A = 3\pi$$

The exact function is:

$$y(x) = 3\pi e^{-x} + 5\pi$$

Finally, evaluate the function at $x = 1$:

$$y(1) = 3\pi e^{-1} + 5\pi = \pi(3e^{-1} + 5)$$

Friday 10 July 2026

5.1 Medium Differential

Bernoulli Differential Equation

By considering a suitable substitution, solve the differential equation

$$\frac{dy}{dx} = \frac{y}{x} - \frac{1}{y}$$

given that the curve passes through $(2, 2)$. Hence, find the value of $x - y$ given that $x = 8$ and $y > 0$.

Solution: We rewrite the ODE in the standard Bernoulli form:

$$y' - \frac{1}{x}y = -y^{-1}$$

Let $v = y^2$. Differentiating with respect to x gives $v' = 2yy'$. Multiplying the entire ODE by $2y$ yields:

$$2yy' - \frac{2}{x}y^2 = -2 \implies v' - \frac{2}{x}v = -2$$

This is a first-order linear ODE. The integrating factor is $I(x) = e^{\int -2/x dx} = e^{-2 \ln x} = x^{-2}$. Multiplying the linear equation by the integrating factor:

$$x^{-2}v' - 2x^{-3}v = -2x^{-2} \implies (vx^{-2})' = -2x^{-2}$$

Integrating both sides with respect to x :

$$vx^{-2} = \int -2x^{-2} dx = \frac{2}{x} + C \implies v = 2x + Cx^2$$

Substituting back $v = y^2$, our general solution is:

$$y^2 = 2x + Cx^2$$

We use the initial condition $(2, 2)$ to find C :

$$2^2 = 2(2) + C(2^2) \implies 4 = 4 + 4C \implies C = 0$$

The specific curve is $y^2 = 2x$. We want to evaluate $x - y$ when $x = 8$. Since $y > 0$:

$$y^2 = 2(8) = 16 \implies y = 4$$

$$x - y = 8 - 4 = 4$$

5.2 Hard Differential

Clairaut's Equation and Parametric Curves

Let curve ζ be a special solution of the differential equation:

$$\left(y - \frac{dy}{dx}x\right) \sec\left(\frac{dy}{dx}\left(\sin\left(\frac{dy}{dx}\right) - 5\right)\right) = 1$$

Write parametric equations of this curve $x = x(t), y = y(t)$ where t is the angular coefficient of the tangent line to it. Find the exact value of $27x(0) + 99y(0)$. Assume continuity over the interval.

Solution: Let $p = \frac{dy}{dx}$. The equation can be rewritten as:

$$(y - px) \sec(p(\sin p - 5)) = 1 \implies y = px + \cos(p(\sin p - 5))$$

This is a Clairaut's equation of the form $y = px + f(p)$, where $f(p) = \cos(p(\sin p - 5))$. The general solution consists of straight lines $y = Cx + f(C)$, while the singular solution (the envelope of these lines, which corresponds to the special curve ζ) is found by differentiating with respect to x :

$$p = p + x \frac{dp}{dx} + f'(p) \frac{dp}{dx} \implies \frac{dp}{dx}(x + f'(p)) = 0$$

For the singular solution, we set $x = -f'(p)$. Since t is the angular coefficient of the tangent, $t = p$. First, we find $f'(t)$:

$$f(t) = \cos(t \sin t - 5t)$$

$$f'(t) = -\sin(t \sin t - 5t) \cdot \frac{d}{dt}(t \sin t - 5t) = -\sin(t \sin t - 5t)(\sin t + t \cos t - 5)$$

This immediately gives us the parametric equation for $x(t)$:

$$x(t) = -f'(t) = \sin(t \sin t - 5t)(\sin t + t \cos t - 5)$$

The corresponding $y(t)$ is found using $y = tx + f(t)$:

$$y(t) = t \sin(t \sin t - 5t)(\sin t + t \cos t - 5) + \cos(t \sin t - 5t)$$

To find $27x(0) + 99y(0)$, we evaluate the parametric equations at $t = 0$:

$$x(0) = \sin(0)(0 + 0 - 5) = 0$$

$$y(0) = 0 + \cos(0) = 1$$

Therefore, the final evaluated sum is:

$$27(0) + 99(1) = 99$$